On Cramér-Rao lower bounds with random equality constraints

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General deterministic estimation problem

Model [Van Trees, 1968]

- parameter space $\Theta_d \subset \mathbb{R}^P$;

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$\rightarrow$ "standard" problem if closed-form of $p(\mathbf{x}; \boldsymbol{\theta})$ available [Kbayer et al., 2017].

Deterministic performance bounds

Cramér-Rao bound (CRB)

[Fréchet, 1943 ; Cramér, 1946 ; Rao, 1945]

- easy to compute ;
- lowest bound on the MSE of unbiased estimators ;
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Constrained CRB (CCRB)

[Gorman and Hero, 1990 ; Marzetta, 1993 ; Stoica et al., 1998-2001 ; Menni et al., 2012]

- $\Theta_d \subset \mathbb{R}^P$: results from **deterministic** equality constraints ;
- versatile for performance analysis and design.

Background on CRB

- Local unbiasedness : $E_{\boldsymbol{\theta}+d\boldsymbol{\theta}}\left[\widehat{\boldsymbol{\theta}}\right] = \boldsymbol{\theta} + d\boldsymbol{\theta} + \mathbf{o}\left(\|d\boldsymbol{\theta}\|\right)$;
  - $\rightarrow$ Taylor expansion.
- Linear constraints :
  $E_{\mathbf{x};\boldsymbol{\theta}}\left[\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}\right] = \mathbf{0},\ E_{\mathbf{x};\boldsymbol{\theta}}\left[(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta})\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}^T}\right] = \mathbf{I}$;
- Fisher information matrix (FIM) :
  $\mathbf{F}\left(\boldsymbol{\theta}\right) = E_{\mathbf{x};\boldsymbol{\theta}}\left[\left(\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}^T}\right)^T \frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}^T}\right]$;
  - $\rightarrow$ $\mathbf{CRB}\left(\boldsymbol{\theta}\right) = \mathbf{F}^{-1}\left(\boldsymbol{\theta}\right) \leq E_{\mathbf{x};\boldsymbol{\theta}}\left[(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta})(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta})^T\right]$.

Equality constraints involving a random parameter

$\boldsymbol{\theta}_r \in \Theta_r \subset \mathbb{R}^{P_r}$: **random** vector parameter.

Probabilistic mapping [Miller et al., 1978 ; D'Andrea et al., 1994]

- i) $\boldsymbol{\theta} \to \boldsymbol{\theta}_r$, ii) $(\boldsymbol{\theta}, \boldsymbol{\theta}_r) \to \mathbf{x}$;

$$p\left(\mathbf{x}, \boldsymbol{\theta}_r; \boldsymbol{\theta}\right) = p\left(\mathbf{x} | \boldsymbol{\theta}_r; \boldsymbol{\theta}\right) p\left(\boldsymbol{\theta}_r; \boldsymbol{\theta}\right).$$

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Aim

derive a CR-like bound : randomly constrained CRB (RCCRB).

Non-standard estimation problem

If $p(\mathbf{x};\boldsymbol{\theta}) = \int_{\Theta_r} p(\mathbf{x},\boldsymbol{\theta}_r;\boldsymbol{\theta})\, d\boldsymbol{\theta}_r$ unknown :

$\rightarrow$ non-standard estimation problem [Kbayer et al., 2017].

$$E_{\mathbf{x};\boldsymbol{\theta}}\left[(\widehat{\boldsymbol{\theta}}-\boldsymbol{\theta})(\widehat{\boldsymbol{\theta}}-\boldsymbol{\theta})^T\right] = E_{\boldsymbol{\theta}_r;\boldsymbol{\theta}}\left[E_{\mathbf{x}|\boldsymbol{\theta}_r;\boldsymbol{\theta}}\left[(\widehat{\boldsymbol{\theta}}-\boldsymbol{\theta})(\widehat{\boldsymbol{\theta}}-\boldsymbol{\theta})^T\right]\right].$$

Random equality constraints

Random equality constraints : $\mathbf{f}_{\boldsymbol{\theta}_r}(\boldsymbol{\theta}) = \mathbf{0} \in \mathbb{R}^K \subseteq \mathcal{C}_{\boldsymbol{\theta}_r} \subset \Theta_d$;

$\rightarrow$ $\mathbf{J}_{\boldsymbol{\theta}_r}(\boldsymbol{\theta}) = \left[\partial \mathbf{f}_{\boldsymbol{\theta}_r}(\boldsymbol{\theta}) / \partial \boldsymbol{\theta}^T\right] \in \mathbb{R}^{K \times P}$: full row rank K.

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$$\left\{ \begin{array}{c} \mathbf{f}_{\boldsymbol{\theta}_r}(\boldsymbol{\theta}) = \mathbf{0} \\ \frac{\partial \mathbf{f}_{\boldsymbol{\theta}_r}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}^T} d\boldsymbol{\theta} = \mathbf{0} \end{array} \right. \Leftrightarrow \left\{ \begin{array}{c} \mathbf{0} = \mathbf{f}_{\boldsymbol{\theta}_r}(\boldsymbol{\theta}) \\ d\boldsymbol{\theta} = \mathbf{U}_{\boldsymbol{\theta}_r}(\boldsymbol{\theta})\, d\lambda \end{array} \right. \text{ when } \|d\boldsymbol{\theta}\| \rightarrow 0.$$

$\rightarrow$ $\mathbf{U}_{\boldsymbol{\theta}_r}(\boldsymbol{\theta})$: basis for $\ker \mathbf{J}_{\boldsymbol{\theta}_r}(\boldsymbol{\theta})$ [Menni et al., 2012].

Conditionally to the value of $\boldsymbol{\theta}_r$

Desirable properties (local unbiasedness)

$$E_{\mathbf{x}|\boldsymbol{\theta}_r;\boldsymbol{\theta}+d\boldsymbol{\theta}}\left[\widehat{\boldsymbol{\theta}}\right] = \boldsymbol{\theta} + d\boldsymbol{\theta} + \mathbf{o}_{\boldsymbol{\theta}_r}\left(\|d\boldsymbol{\theta}\|\right),$$

$$E_{\mathbf{x}|\boldsymbol{\theta}_r+d\boldsymbol{\theta}_r;\boldsymbol{\theta}}\left[\widehat{\boldsymbol{\theta}}\right] = \boldsymbol{\theta} + \boldsymbol{o}\left(\|d\boldsymbol{\theta}_r\|\right), \ \forall \boldsymbol{\theta} \in \mathcal{C}_{\boldsymbol{\theta}_r}.$$

$$\Rightarrow \left\{ \begin{array}{r} E_{\mathbf{x}|\boldsymbol{\theta}_r;\boldsymbol{\theta}}\left[\left(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}\right)\left(\mathbf{U}_{\boldsymbol{\theta}_r}^T\left(\boldsymbol{\theta}\right)\frac{\partial \ln p(\mathbf{x}|\boldsymbol{\theta}_r;\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right)^T\right] = \mathbf{U}_{\boldsymbol{\theta}_r}\left(\boldsymbol{\theta}\right), \\ E_{\mathbf{x}|\boldsymbol{\theta}_r;\boldsymbol{\theta}}\left[\left(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}\right)\frac{\partial \ln p(\mathbf{x}|\theta_r;\boldsymbol{\theta})}{\partial \boldsymbol{\theta}_r^T}\right] = \mathbf{0}. \end{array} \right.$$

Expression of the RCCRB

Conditional CCRB

$$\mathbf{CCRB}_{\boldsymbol{\theta}_r}(\boldsymbol{\theta}) = \mathbf{U}_{\boldsymbol{\theta}_r}(\boldsymbol{\theta}) \left(\mathbf{U}_{\boldsymbol{\theta}_r}^T(\boldsymbol{\theta}) \, \mathbf{CRB}_{\boldsymbol{\theta}_r}^{-1}(\boldsymbol{\theta}) \, \mathbf{U}_{\boldsymbol{\theta}_r}(\boldsymbol{\theta}) \right)^{-1} \mathbf{U}_{\boldsymbol{\theta}_r}^T(\boldsymbol{\theta}) \, .$$

Assumptions :

- $\widehat{\boldsymbol{\theta}} \triangleq \widehat{\boldsymbol{\theta}}(\mathbf{x})$: locally unbiased cond. to $\boldsymbol{\theta}_r$;
- $\widehat{\boldsymbol{\theta}} \in \mathcal{C}_{\boldsymbol{\theta}_r} \subset \Theta_d$

RCCRB

$$\mathbf{RCCRB}(\boldsymbol{\theta}) = E_{\boldsymbol{\theta}_r;\boldsymbol{\theta}}\left[\mathbf{CCRB}_{\boldsymbol{\theta}_r}(\boldsymbol{\theta})\right],$$

$$E_{\mathbf{x}|\boldsymbol{\theta}}\left[(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta})(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta})^T\right] \geq \mathbf{RCCRB}(\boldsymbol{\theta}) \, .$$

Additional remarks

If $p\left(\mathbf{x}, \boldsymbol{\theta}_r; \boldsymbol{\theta}\right) = p\left(\mathbf{x}; \boldsymbol{\theta}\right) p\left(\boldsymbol{\theta}_r; \boldsymbol{\theta}\right)$

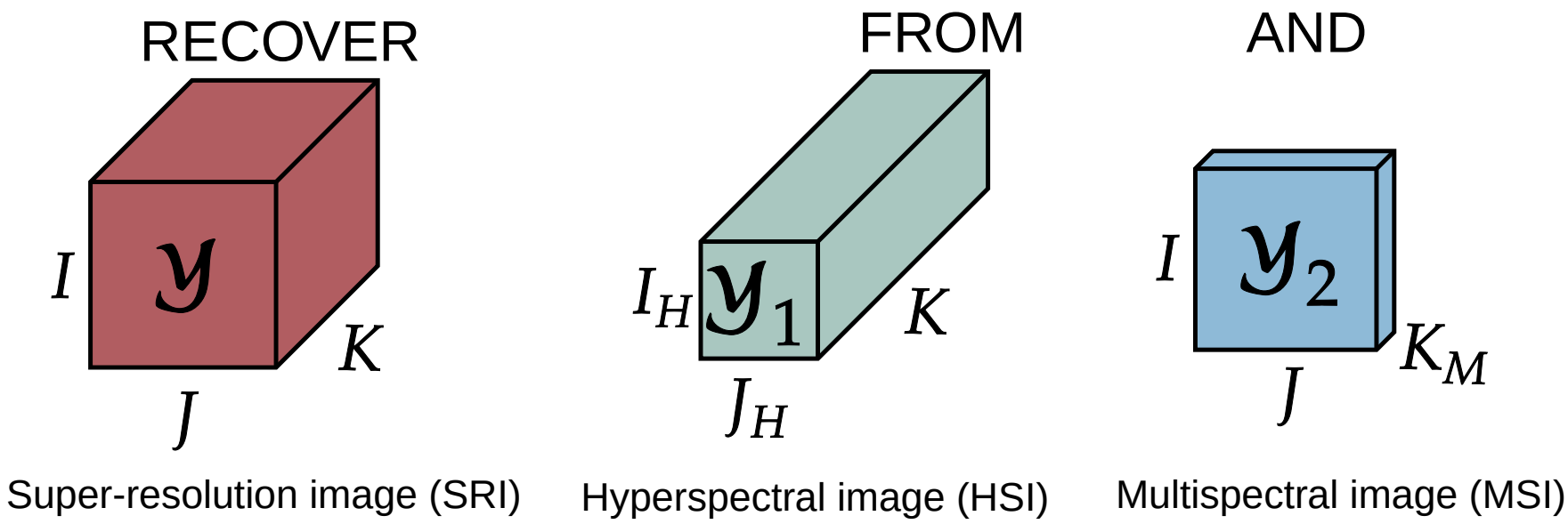
$$p\left(\mathbf{x}|\boldsymbol{\theta}_r; \boldsymbol{\theta}\right) = p\left(\mathbf{x}; \boldsymbol{\theta}\right) \Rightarrow \mathbf{CRB}_{\boldsymbol{\theta}_r}^{-1}\left(\boldsymbol{\theta}\right) = \mathbf{F}_{\boldsymbol{\theta}_r}\left(\boldsymbol{\theta}\right) = \mathbf{F}\left(\boldsymbol{\theta}\right),$$

$$\mathbf{RCCRB}\left(\boldsymbol{\theta}\right) = E_{\boldsymbol{\theta}_r;\boldsymbol{\theta}}\left[\mathbf{U}_{\boldsymbol{\theta}_r}\left(\boldsymbol{\theta}\right)\left(\mathbf{U}_{\boldsymbol{\theta}_r}^T\left(\boldsymbol{\theta}\right)\mathbf{F}\left(\boldsymbol{\theta}\right)\mathbf{U}_{\boldsymbol{\theta}_r}\left(\boldsymbol{\theta}\right)\right)^{-1}\mathbf{U}_{\boldsymbol{\theta}_r}^T\left(\boldsymbol{\theta}\right)\right].$$

If the constraints are non-random :

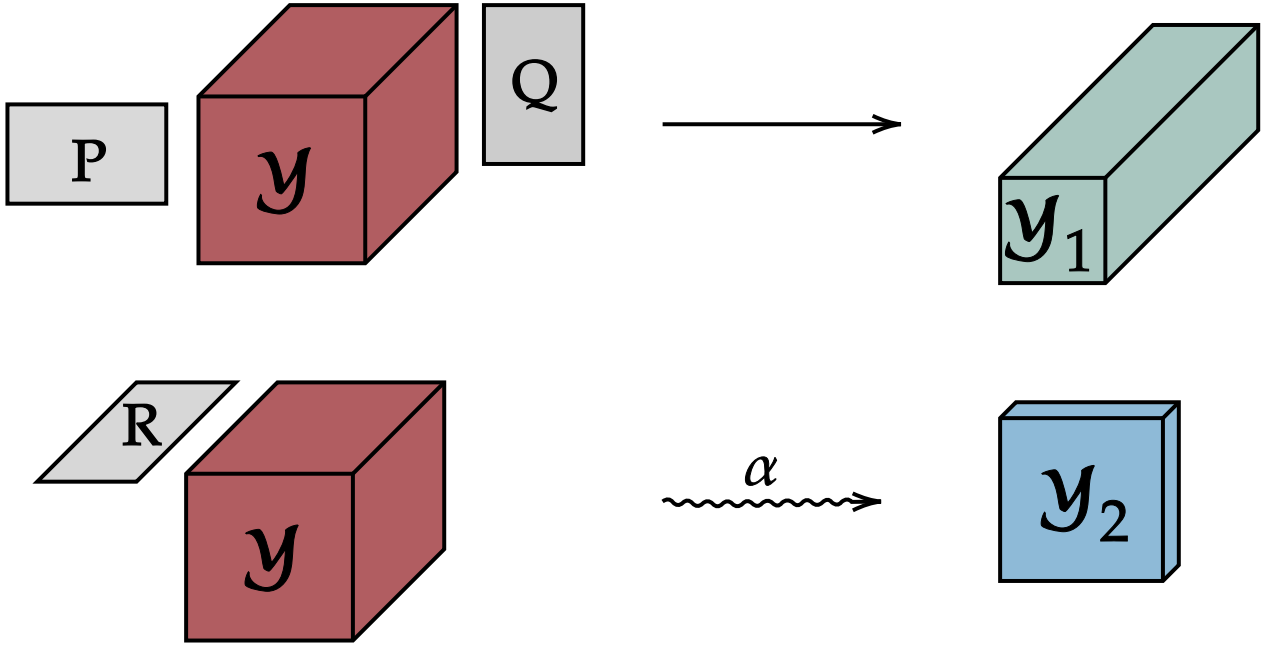
$$\mathbf{RCCRB}\left(\boldsymbol{\theta}\right) = \mathbf{CCRB}\left(\boldsymbol{\theta}\right).$$

Hyperspectral super-resolution problem



- $\mathcal{Y}_1$: low spatial resolution $(I_H < I, J_H < J)$;
- $\mathcal{Y}_2$: low spectral resolution $(K_M < K)$.

HSR degradation model



$$\begin{cases} \mathcal{Y}_1 &= \mathcal{Y} \bullet_1 \boldsymbol{P} \bullet_2 \boldsymbol{Q}, \\ \mathcal{Y}_2 &= \mathcal{Y} \bullet_3 \alpha \boldsymbol{R}. \end{cases}$$

- $\boldsymbol{P}$, $\boldsymbol{Q}$, $\boldsymbol{R}$: full row rank degradation matrices ;
- α : **unknown** variation in illumination conditions [Camacho et al., 2019].

CP model

$$\begin{cases} \boldsymbol{\mathcal{Y}}_1 &= [\![\boldsymbol{A}_1, \boldsymbol{B}_1, \boldsymbol{C}_1]\!] + \boldsymbol{\mathcal{E}}_1, \quad \boldsymbol{\mathcal{E}}_1 \sim \boldsymbol{\mathcal{N}}(0, \sigma_1^2 \boldsymbol{I}) \\ \boldsymbol{\mathcal{Y}}_2 &= [\![\boldsymbol{A}_2, \boldsymbol{B}_2, \boldsymbol{C}_2]\!] + \boldsymbol{\mathcal{E}}_2, \quad \boldsymbol{\mathcal{E}}_2 \sim \boldsymbol{\mathcal{N}}(0, \sigma_2^2 \boldsymbol{I}) \end{cases}$$

subject to $\boldsymbol{A}_1 = \boldsymbol{P}\boldsymbol{A}_2, \boldsymbol{B}_1 = \boldsymbol{Q}\boldsymbol{B}_2$, and $\boldsymbol{C}_2 = \alpha \boldsymbol{R}\boldsymbol{C}_1$.

STEREO - criterion to minimize [Kanatsoulis et al., 2018]

$$\|\boldsymbol{\mathcal{Y}}_1 - [\![\boldsymbol{A}_1, \boldsymbol{B}_1, \boldsymbol{C}_1]\!]\|_F^2 + \lambda \|\boldsymbol{\mathcal{Y}}_2 - [\![\boldsymbol{A}_2, \boldsymbol{B}_2, \boldsymbol{C}_2]\!]\|_F^2,$$
$$\text{subject to } \boldsymbol{A}_1 = \boldsymbol{P}\boldsymbol{A}_2, \boldsymbol{B}_1 = \boldsymbol{Q}\boldsymbol{B}_2, \boldsymbol{C}_2 = \boldsymbol{R}\boldsymbol{C}_1.$$

- $\lambda = \frac{\sigma_1^2}{\sigma_2^2} \rightarrow$ Log-likelihood criterion ;
  $\rightarrow$ STEREO = MLE.
- identifiability of $\boldsymbol{\mathcal{Y}}_1$: not required ! [Kanatsoulis et al., 2018]

Model parameters

$$\boldsymbol{\psi}_1 = \mathrm{vec}\{\boldsymbol{C}_1\} \in \mathbb{R}^{KN\times 1}, \qquad \boldsymbol{\phi}_1 = \begin{bmatrix}\mathrm{vec}\{\boldsymbol{A}_1\}\\ \mathrm{vec}\{\boldsymbol{B}_1\}\end{bmatrix} \in \mathbb{R}^{(I_H+J_H)N\times 1},$$

$$\boldsymbol{\psi}_2 = \mathrm{vec}\{\boldsymbol{C}_2\} \in \mathbb{R}^{K_M N\times 1}, \quad \boldsymbol{\phi}_2 = \begin{bmatrix}\mathrm{vec}\{\boldsymbol{A}_2\}\\ \mathrm{vec}\{\boldsymbol{B}_2\}\end{bmatrix} \in \mathbb{R}^{(I+J)N\times 1}.$$

- HSI : only degraded *spatially* ;
- MSI : only degraded *spectrally* ;
- $\boldsymbol{\theta}^\mathsf{T} = \begin{bmatrix}\boldsymbol{\psi}_1^\mathsf{T} & \boldsymbol{\phi}_1^\mathsf{T} & \boldsymbol{\psi}_2^\mathsf{T} & \boldsymbol{\phi}_2^\mathsf{T}\end{bmatrix}$, $\boldsymbol{\theta}_r = \alpha$.

Random equality constraints

Previous works

- Standard CRB [Ren et al., 2016] ;
- Deterministic CCRB, $\boldsymbol{\mathcal{Y}}_1, \boldsymbol{\mathcal{Y}}_2$ identifiable : [Ren et al., 2016 ; Cabral Farias et al., 2016 ; Prévost et al., 2019] ;

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Random equality constraints

$$\mathbf{f}_{\boldsymbol{\theta}_r}(\boldsymbol{\theta}) = \begin{cases} \boldsymbol{\psi}_2 - (\boldsymbol{I} \boxtimes \alpha \boldsymbol{R})\boldsymbol{\psi}_1 = \mathbf{0}, \\ \boldsymbol{\phi}_1 - \begin{bmatrix} \boldsymbol{I} \boxtimes \boldsymbol{P} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I} \boxtimes \boldsymbol{Q} \end{bmatrix} \boldsymbol{\phi}_2 = \mathbf{0}. \end{cases}$$

Simulations setup

- $I = J = 18$, $K = 16$, $I_H = J_H = 6$, $K_M = 8$, $N = 3$;
- $\boldsymbol{A}_i, \boldsymbol{B}_i, \boldsymbol{C}_i$: i.i.d. real Gaussian variables ;
- $\boldsymbol{P}, \boldsymbol{Q}, \boldsymbol{R}$: generated from Wald's protocol ;
- ALS + STEREO (500 iter. max.) : MSE over 500 realizations ;
- SNR on $\boldsymbol{\mathcal{Y}}_1$: 5 to 60dB ; SNR on $\boldsymbol{\mathcal{Y}}_2$: 20dB.

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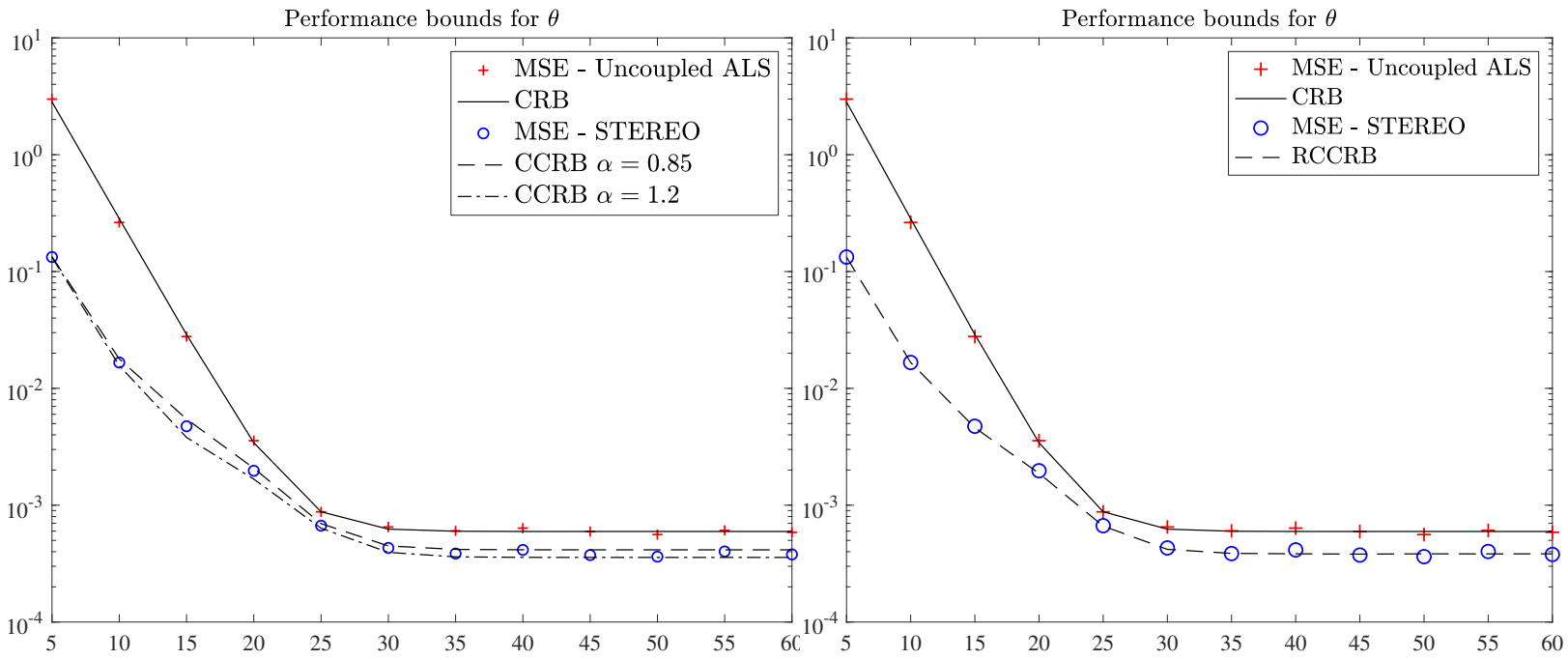
- Conditional CCRB : $\alpha = 0.85$ and $\alpha = 1.2$ [Menni et al., 2012] ;
- $\alpha \sim \boldsymbol{\mathcal{N}}(1, 0.08)$;
- RCCRB : average over 200 realizations.

Results


Performance bounds for θ
MSE - Uncoupled ALS
CRB
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MSE - STEREO
RCCRB

Conclusion

- RCCRB : novel bound for random equality constraints ;
- Tighter than expectation of conditional CCRB ;
- Illustrates well the HSR problem with variations in illumination ;
- Does not require identifiability of the HSI.

Thank you for your attention !